STAT 325 Design of Experiments
Homework 4
2/19/26
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3.4. A computer ANOVA output is shown below. Fill in the blanks. You may give bounds on the P-value.

One-way ANOVA

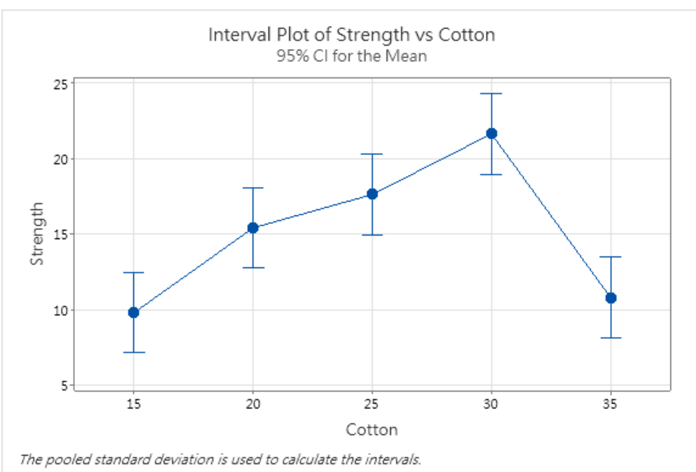
Source	DF	SS	MS	F	P
Factor	?	?	246.93	?	?
Error	25	186.53	?		
Total	29	1174.24			

Source	DF	SS	MS	F	P
Factor	4	987.71	246.93	33.09521257	<0.001
Error	25	186.53	7.4612		
Total	29	1174.24			

Factor DF = 29 - 25 = 4
Factor SS = 1174.24- 186.53= 987.71
Error MS = 186.53/25 = 7.4612
F = 246.93/7.4612 = 33.095
P = Fcdf (Lower = F_0 = 33.09521, Upper = 999, df_num = 4, df_den= 25)

3.10. A product developer is investigating the tensile strength of a new synthetic fiber that will be used to make cloth for men's shirts. Strength is usually affected by the percentage of cotton used in the blend of materials for the fiber. The engineer conducts a completely randomized experiment with five levels of cotton content and replicates the experiment five times. The data are shown in the following table.

Cotton Weight Percent	Observations				
15	7	7	15	11	9
20	12	17	12	18	18
25	14	19	19	18	18
30	19	25	22	19	23
35	7	10	11	15	11



Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Cotton	4	475.8	118.940	14.76	0.000
Error	20	161.2	8.060		
Total	24	637.0			

(a) Is there evidence to support the claim that cotton content affects the mean tensile strength? Use 0.05.

Using $\alpha = 0.05$ we see that the P-value is near 0 and we reject the null hypothesis where the null is that all the treatment means or in this case the cotton weight percentages means are the same. So we rejected that the means are the same and also we can visually see that if we plotted all the means strengths by from each cotton weight percentage groups, they are clearly different from each other as well by a fair margin.

(b) Use the Fisher LSD method to make comparisons between the pairs of means. What conclusions can you draw?

(c) Analyze the residuals from this experiment and comment on model adequacy.

3.44. Suppose that four normal populations have means of $\mu_1 = 50$, $\mu_2 = 60$, $\mu_3 = 50$, and $\mu_4 = 60$. How many observations should be taken from each population so that the probability of rejecting the null hypothesis of equal population means is at least 0.90? Assume that $\alpha = 0.05$ and that a reasonable estimate of the error variance is $\sigma^2 = 25$.